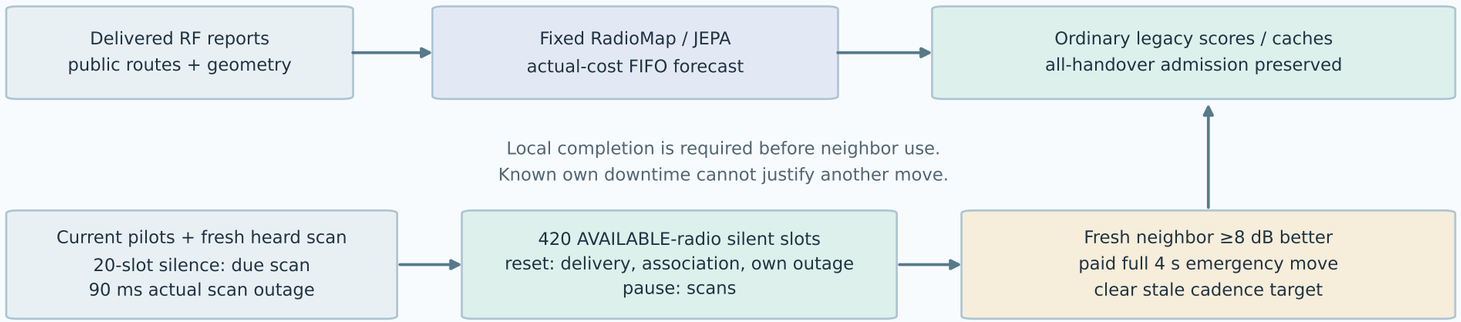
Available-radio evidence, actual recovery and paid completion

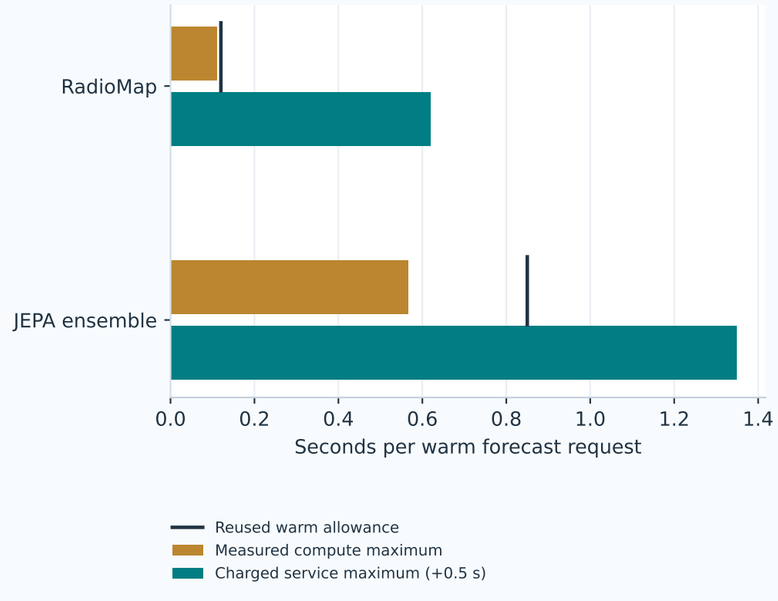
A four-second-only, independent prospective study. Own handover downtime resets the watchdog; scans pause its evidence counter.

Native packet and action traces are descriptive diagnostics; registered statistical and engineering decisions remain in the primary figure.

A Preserve ordinary policy; gate emergency action on available-radio evidence



B Warm forecast work and service charge

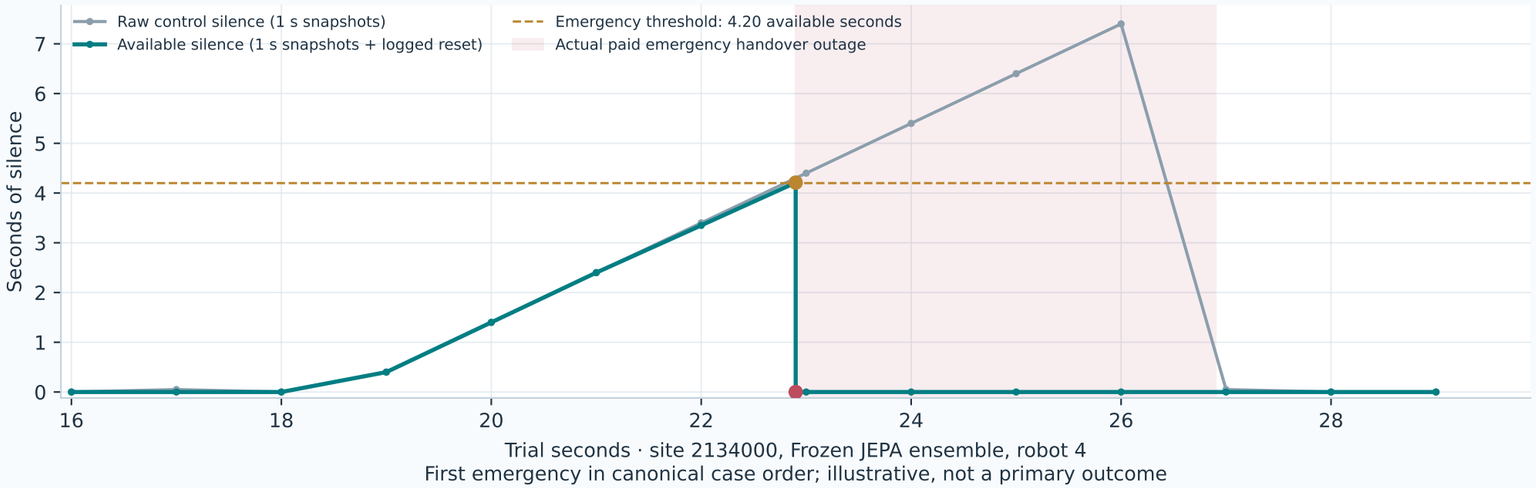


C Actual recovery, sensing and audit counts

Family	Watchdog moves	Due scans while silent	All moves legacy → repair
Map	0.88	14.00	61.1 → 62.2
JEPA	1.12	12.88	54.9 → 55.6

Confirmation cases: 32; source / paid / causal: PASS
Actual jobs: 1,920; profile-exceeded cases: 0
Local watchdog checks: 7353
Largest consumed forecast age: 17.98 s
Exact offline replays: 4 / 4
Fresh-cost repetitions: 4; changed availability: 0
Fresh max absolute miss change: 0
Counts above: means per 100-robot, 60 s trial.

D An actual local evidence counter beside raw silence



Measured history-to-forecast issue work determines actual FIFO completion. Startup/loading is separate; shared +0.5 s is a scheduling assumption. Ordinary admission/scoring CPU is not separately scheduled.
A move resets available silence to zero; delivery / association / own handover reset it, scans pause it. The full 4 s handover and 90 ms scan costs are paid.
Offline replay retains the original paid schedule; fresh-cost repetitions measure timing sensitivity without extra sites. Shared procedural CAD, finite trials and warm history do not establish physical reliability.

PRIOR AGGRESSIVE 4 s REPAIR FAILED DEVELOPMENT: RadioMap +5.505 pp · JEPA +4.748 pp. That result remains unchanged.

This independent cost-aware study uses disjoint new site IDs. Prior failed study: jepa_repair/figures/confirmation.svg